

Class 4 Deductive Reasoning & Valid Arguments

When constructing mathematical proofs, we need to string together several sentences called premises or assumptions and then use some rule of deduction to draw a conclusion.

Example

An _____ is a list of sentences called
_____ and a _____

An argument is called valid if

If we represent all the premises (assumptions) by $A_1, A_2, \dots, A_n$ and the conclusion by C , we say the argument $A_1, A_2, \dots, A_n \dots C$ is valid if we never have all the A_i 's being true and C false. This means the sentence $(A_1 \wedge A_2 \wedge \dots \wedge A_n) \rightarrow C$ is always true if the argument is valid.

Example of a valid argument

Modus Ponens

Valid Rules of Deduction

Formal Name	Rule of Deduction
Modus Ponens	$p \rightarrow q, p \therefore q$
Modus Tollens	$p \rightarrow q, \sim q \therefore \sim p$
Double Negation	$\sim(\sim p) \therefore p$ $p \therefore \sim(\sim p)$
Conjunctive Simplification	$p \wedge q \therefore p$ $p \wedge q \therefore q$
Conjunctive Addition	$p, q \therefore p \wedge q$ $p, q \therefore q \wedge p$
Disjunctive Syllogism	$p \vee q, \sim p \therefore q$ $p \vee q, \sim q \therefore p$
Disjunctive Addition	$p \therefore p \vee q$ $q \therefore p \vee q$
Hypothetical Syllogism	$p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$
Dilemma	$p \vee q, p \rightarrow r, q \rightarrow r \therefore r$
Contradiction	$(\sim p) \rightarrow [q \wedge (\sim q)] \therefore p$
Logical Equivalence if $B \equiv C$	$B \therefore C$
DeMorgan's Law	$\sim(p \wedge q) \therefore (\sim p) \vee (\sim q)$ $(\sim p) \vee (\sim q) \therefore \sim(p \wedge q)$ $\sim(p \vee q) \therefore (\sim p) \wedge (\sim q)$ $(\sim p) \wedge (\sim q) \therefore \sim(p \vee q)$

Prove Disjunctive Syllogism
is a valid rule of deduction

Use a truth table to prove the following argument is not valid.

$$p \rightarrow q, \sim p \therefore \sim q$$

If there are several variables, the truth table may be so big you don't want to construct the whole thing.

Alternate method to prove an argument is invalid

Example: Prove $(p \wedge q) \rightarrow r, \sim q \therefore r$ is not valid.

Example: Use rules of deduction to prove the argument

$(A \vee B) \rightarrow C, B \therefore C$ is valid.

Example: Use rules of deduction to prove the argument

$A \vee (\sim C), B \wedge E, B \rightarrow (C \wedge D) \therefore A$ is valid.